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Original Contribution

Impact of Changes to Concealed-Carry Weapons Laws on Fatal and Nonfatal Violent Crime, 1980–2019

Mitchell L. Doucette*, Alexander D. McCourt, Cassandra K. Crifasi, and Daniel W. Webster

* Correspondence to Dr. Mitchell L. Doucette, Center for Gun Violence Solutions, Department of Health Policy and Management, Johns Hopkins Bloomberg School of Public Health, 615 N. Wolfe Street, Baltimore, MD 21205 (e-mail: mdoucet3@jhu.edu).

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The United States faces rapidly rising rates of violent crime committed with firearms. In this study, we sought to estimate the impact of changes to laws that regulate the concealed carrying of weapons (concealed-carry weapons (CCW) laws) on violent crimes committed with a firearm. We used augmented synthetic control models and random-effects meta-analyses to estimate state-specific effects and the average effect of adopting shall-issue CCW permitting laws on rates of 6 violent crimes: homicide with a gun, homicide by other means, aggravated assault with a gun, aggravated assault with a knife, robbery with a gun, and robbery with a knife. The average effects were stratified according to the presence or absence of several shall-issue permit provisions. Adoption of a shall-issue CCW law was associated with a 9.5% increase in rates of assault with a firearm during the first 10 years after law adoption and was associated with an 8.8% increase in rates of homicide by other means. When shall-issue laws allowed violent misdemeanants to acquire CCW permits, the laws were associated with higher rates of gun assaults. It is likely that adoption of shall-issue CCW laws has increased rates of nonfatal violent crime committed with firearms. Harmful effects of shall-issue laws are most clear when provisions intended to reduce risks associated with civilian gun-carrying are absent.

concealed-carry laws; firearms; firearm violence; guns; gun violence; health policy evaluation; synthetic control method; weapons

Abbreviations: ASCM, augmented synthetic control modeling; ASCM+FE, augmented synthetic control modeling with fixed effects; ATT, average treatment effect in the treated; CCW, concealed-carry weapons; CDC, Centers for Disease Control and Prevention; CI, confidence interval; REMA, random-effects meta-analysis; RMSE, root mean square error; SCM, synthetic control modeling; SCM+FE, synthetic control modeling with fixed effects; UCR, Uniform Crime Reporting.

Editor's note: *An invited commentary on this article and the authors' response will appear in an upcoming issue.*

Though some states allow the open carrying of firearms, laws that regulate the carrying of concealed firearms have been intensely debated for the last few decades. During the past 40 years, many states have removed restrictions on concealed gun-carrying. In 1980, 21 states did not permit civilians to carry a concealed firearm under any circumstances. In 2021, 21 states did not require any special vetting or licensing of individuals who wished to carry concealed firearms in public, referred to as "permitless" concealed-carry weapons (CCW) laws, and an additional 21 states had what is known as "shall-issue" CCW permitting laws, which place relatively modest requirements on legal gun owners for carrying their firearms concealed outside of the home. The remaining 8 states and the District of Columbia had "may-issue" CCW laws, which provide law enforcement agencies with considerable discretion over who they issue a license to carry weapons to and often require applicants to prove that they have a "good cause" or special cause for needing to carry a concealed gun. Although these 8 states have maintained may-issue policies, many of these laws are under scrutiny as part of ongoing litigation (1).

The changing landscape of state laws governing concealed gun-carrying over the past 40 years has provided researchers an opportunity to estimate the relationship

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between these laws and public safety. Indeed, no other type of gun law has been studied as frequently with a wide array of methods and findings (2). Early research on CCW laws by Lott and Mustard (3) suggested that movement from no-issue or may-issue laws to shall-issue laws led to reductions in violent crime. The "more guns, less crime" hypothesis that was foundational to this work posited that if more civilians were legally carrying firearms in public, criminals would be deterred from engaging in crime, as they would not know whether a potential target was armed. The study, however, contained serious flaws previously noted in the literature (4, 5). Despite these flaws, this hypothesis and perceived deterrent benefits have been used to argue for lowering restrictions on concealed gun-carrying.

Recent research (2, 5–14) estimated the impacts of shall-issue laws on violent crime outcomes, with additional years of data and statistical techniques that addressed problems present in Lott and Mustard's research. Many studies that sought to estimate the association between CCW laws and violent crime have been described and critiqued elsewhere (2). Findings have been somewhat mixed; however, many of the most recent studies have found evidence that adopting a right-to-carry law is associated with increases in violent crime, including homicides (5–14). The majority of these studies used panel regression to estimate the impact of right-to-carry laws on violent outcomes as Lott and Mustard did (3), and most addressed weaknesses in the above Lott and Mustard study. Moreover, when studies examined the relationship between CCW laws and violent crime, they used total nonfatal violent crime as the outcome measure rather than disaggregated violent crimes committed with firearms.

Some shall-issue laws have certain restrictions or requirements for obtaining a license to carry a concealed firearm, and these provisions have not been examined in prior studies. These provisions are likely to influence the number of licenses issued, how well individuals with histories of violence are screened out, and the ability of the person applying for a license to carry to safely handle a firearm. In this study, we estimated the association between violent crime and moving from a may- or no-issue CCW law to a shall-issue CCW law within strata defined by whether: 1) violent misdemeanants are ineligible to receive a license to carry; 2) officials can deny applicants based on a history of dangerous behavior; 3) the law has a "suitability" clause that allows officials to deny a license to someone deemed to be of questionable character; and 4) the law requires firearm safety training that includes applicants' engaging in some form of live-fire training. The first 3 restrictive provisions could reduce the number of individuals at elevated risk for committing acts of violence who can legally carry concealed firearms, and the fourth could potentially prevent individuals who are insufficient at handling a firearm safely from carrying a gun and reduce the overall number of people licensed to carry, because the safety tests require time and money (15).

The current study examined relationships between the adoption of shall-issue laws and rates of disaggregated nonfatal and fatal violent crime committed with a firearm and other means. Further, we sought to understand whether the impact of shall-issue law adoption varies based on the presence or absence of provisions intended to limit risks associated with civilian gun-carrying. We hypothesized that moving to a shall-issue CCW law from a may-issue/no-issue CCW law increased rates of fatal and nonfatal violent crime at the individual state level and in the aggregate. Further, we hypothesized that CCW permit provisions designed to restrict CCW permits from being issued to risky gun possessors would attenuate violence-producing effects of shall-issue laws.

METHODS

We used a quasi-experimental design to estimate the association between shall-issue law adoption and annual state-level counts of fatal and nonfatal violent crimes. We estimated the average treatment effect in the treated (ATT) for fatal and nonfatal violent crimes related to the changes in CCW laws using a comparative time-series analysis. We identified the average effect of laws weighting state-specific effects based on the inverse of the standard error using random-effects meta-analysis (REMA).

Variables and data sources

This study examined 6 separate outcomes: rates of 1) aggravated assault committed with a gun, 2) robbery committed with a gun, 3) homicide committed with a gun, 4) aggravated assault committed with a knife, 5) robbery committed with a knife, and 6) homicide committed by other means, excluding firearms (nongun homicides). Counts of aggravated assaults and robberies committed with a gun or a knife were ascertained from the Federal Bureau of Investigation's Uniform Crime Reporting (UCR) Program ("Offenses Known and Clearances by Arrest" files) (16). Counts of gun and nongun homicides were obtained from the Centers for Disease Control and Prevention's (CDC) National Center for Health Statistics. We defined firearm homicide using *International Classification of Diseases, Tenth Revision*, codes X93, X94, X95, and *U01.4. Nonfirearm homicides (or homicides by all other means not involving a firearm) were defined using *International Classification of Diseases, Tenth Revision*, codes X85–X92, Y87.1, *U01(.0–.3, .5–.9), and *U02. We accessed the yearly compressed mortality files via data request to the CDC. We aggregated outcomes to the state level and indexed them by year. We obtained population counts by state from the CDC's Web-based Injury Statistics Query and Reporting System (WISQARS) to generate violent crime rates per 100,000 population.

The following covariates were included in each of the statistical models: percentage of the population living in a Metropolitan Statistical Area, percentage of black males aged 15–19 years, percentage of White males aged 15–19 years, percentage of the population living in poverty, and percent unemployed, as well as personal income, rate of ethanol consumption, rate of incarceration, and law enforcement employment per capita (number of law enforcement employees per 100,000 population indexed at the state-year level). Rates of incarceration and police employment were

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extrapolated for 2019. Covariate data were acquired from the US Census Bureau, the Bureau of Labor Statistics, the Bureau of Justice Statistics, the UCR Program, and the National Institute on Alcohol Abuse and Alcoholism. These covariates were chosen on the basis of their demonstrated relationship with violent crime, as described in the literature, and theory: Sociodemographic variables capture factors, including racism and concentrated disadvantage, that affect the distribution of violence; other variables reflect the distribution of policing and incarceration, which affects crime and crime reporting; and alcohol consumption was included to account for the close relationship between alcohol use and violence (7, 17–22).

State laws were identified through searches of each state's legal code. We determined effective dates (dates the law went into effect) through review of each statute's legislative history. We compared our findings with public databases of state gun laws and prior research on concealed-carry laws (5, 23, 24). During our study period, 36 states adopted a shall-issue CCW law as a change from a may-issue/no-issue CCW law. Our analysis focused on evaluating transitions from no-issue or may-issue laws to shall-issue laws and changes in key provisions relevant to reducing risk associated with civilian gun-carrying. The following permit provisions were identified: a live-fire training requirement, discretion to deny whenever an applicant was deemed "unsuitable" (e.g., immoral, unstable), discretion to deny if an applicant had a history of dangerousness, and prohibition based on prior violent misdemeanor convictions. Table 1 provides effective dates for the adoption or repeal of each type of law and permit provision. Within our analysis, we analyzed the presence and absence of each shall-issue law provision separately and analyzed the impact of having no permit provisions and at least 1, 2, or 3 permit provisions.

Analysis

We used a data-driven approach to examine the ATT of adopting a shall-issue CCW law. We compared the performance of 4 model specifications related to the synthetic control method, including synthetic control modeling (SCM), SCM with fixed effects (SCM+FE), augmented synthetic control modeling (ASCM), and ASCM with fixed effects (ASCM+FE). The ASCM method is an extension of the synthetic control method developed by Abadie and Gardeazabal (25). The SCM creates a "synthetic state" for the outcome of interest in the state, with the policy change based on a set of weights among a donor pool of comparison states that were "at risk" of the policy change. Weights are calculated using lagged outcomes and covariate values from the donor states and seek to minimize the error in predicting the observed values of the outcome variable in the treated state during the period prior to the policy change (25–27). The SCM approach then uses those weights to forecast into the post–policy-change period for the synthetic treatment state, allowing a counterfactual comparison (25–28). Specifying fixed effects demeans the lagged outcomes in the pretreatment period prior to the creation of the weighted synthetic control model.

The ASCM method is like the SCM method in that it creates a weighted synthetic state. The synthetic control state is then augmented using a linear regression model, regressing the synthetic state on the treated state. The linear regression model's coefficients are estimated using a ridge estimator, rather than ordinary least squares, introducing the L2-norm penalty term. The use of the ridge estimator can produce more precise standard errors around the true effect value when data suffer from multicollinearity. The L2-norm is fine-tuned using the constant λ, which is selected using leave-one-out cross-validation. This technique works to minimize overfitting of the counterfactual in the pretreatment period. The ridge estimator introduces potentiality for negatively weighted donor states and improves the originally estimated synthetic control pretreatment fit. ASCM has recently been used to examine opioid use (29), firearm violence (30), and air pollution (31). The ASCM method is notably useful when the pool of potential donor states is small.

As part of our model evaluation, we considered the root mean square error (RMSE) during the pretreatment period as well as the RMSE in the 3 years prior to implementation. These measures, and their relative distance from each other, provide an estimation of overall synthetic control model performance, as well as an estimation of how much the synthetic control model diverges from the treatment trend immediately prior to treatment. Synthetic controls with high RMSE divergence in the years prior to implementation may provide biased estimates of the ATT. The ATT is calculated as the average difference between the treated state and the augmented synthetic control state during the postimplementation period. We specified jackknife standard errors for inference (32, 33). We used the R package "augsynth" (34). For all models, the donor pool consisted of the 8 may-issue states with unchanged CCW laws during the study period (California, Connecticut, Delaware, Hawaii, Massachusetts, Maryland, New Jersey, and New York) (Table 1). We did not include late-adopting shall-issue states in our donor pool.

Prior research suggests that the impact of CCW law changes peaks at around 10 years (35). Forecasting counterfactuals for long periods increases risks of confounding factors' biasing estimates of policy impacts. Therefore, each model is restricted to 10 years post–law change. To ensure that the most relevant and recent trends were used to construct the policy counterfactuals, pretreatment training years were restricted to the 10 years prior to law change. Alaska removed its permit provision for carrying a concealed weapon less than 10 years after adopting a shall-issue CCW permitting system. We ended its posttreatment period in the year before the permitless CCW system was enacted. We include model specification data (SCM, SCM+FE, ASCM, and ASCM+FE) for all models, including donor pool weights, covariate balance, and, for ASCM and ASCM+FE, lambda cross-validation.

A priori, we excluded Kansas and Missouri, as well as Florida's nonfatal outcomes, from our analysis due to issues with data reporting and possible confounding. Examination of Florida's participation in the UCR reporting system showed that approximately 95% of law enforcement agencies in the state reported data in the year prior to shall-issue

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Table 1. Legal Landscape Regarding Concealed-Carry Weapons Laws in the United States, 1980–2019[a]

State	Overall CCW Permitting Law			Specific Permitting Requirements			
	May-Issue Permit	Shall-Issue Permit[b]	Permitless Carry	Live Fire	Dangerousness	Suitability	Any Violent Misdemeanor Prohibition for CCW Permit
Alabama		2013[c]			2013	Pre-1980–2013	Pre-1980
Alaska		1994	2003	1994–2003			1994–2003
Arizona		1994	2010				
Arkansas		1994[c]			1995–2021		
California	Pre-1980					Pre-1980	1991
Colorado		2003[c]			2003		
Connecticut	Pre-1980					Pre-1980	1994
Delaware	Pre-1980			1998		Pre-1980	
Florida		1987		2016		Pre-1980–1987	1987
Georgia		1989[c]				Pre-1980	1983
Hawaii	Pre-1980			1995[d]		Pre-1980	1981
Idaho		1990	2016[e]				1990–2016
Illinois		2014[c]		2014	2014		1996
Indiana		Pre-1980[c]				Pre-1980	
Iowa		2011[c]			2011		Pre-1980
Kansas		2007	2015[e]	2007–2015	2007–2015		
Kentucky		1996	2019[e]	1996–2019	1997		
Louisiana		1996			1996		1996
Maine		1981	2015[e]			1981–2015	
Maryland	Pre-1980			2013			1996
Massachusetts	Pre-1980				Pre-1980	Pre-1980	
Michigan		2001		2001		Pre-1980–2001	2001
Minnesota		2003[c]		2003	2003		2003
Mississippi		1991	2016[e]				
Missouri		2004[c]	2017[e]	2004–2017	2004–2007		2004–2017
Montana		1991[c]			1991	Pre-1980	1991
Nebraska		2007		2007			2007
Nevada		1995					1995
New Hampshire		Pre-1980[c]	2017[f]			Pre-1980–2017	
New Jersey	Pre-1980					Pre-1980	
New Mexico		2004		2004		Pre-1980	2004
New York	Pre-1980						Pre-1980
North Carolina		1995		1995			1995
North Dakota		1985	2017[d]	1985–2017	2013–2017		1985
Ohio		2004		2004			2004
Oklahoma		1995	2019[e]	1995–2019			1995–2019
Oregon		1990[c]			1990	Pre-1980–1990	1991
Pennsylvania		1989[c]			1989	Pre-1980	1995
Rhode Island		Pre-1980[c]		Pre-1980		Pre-1980	
South Carolina		1996		Pre-1980			
South Dakota		1985	2019[e]		1985–2019	Pre-1980	

Table continues

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Table 1. Continued

State	Overall CCW Permitting Law			Specific Permitting Requirements			
	May-Issue Permit	Shall-Issue Permit[b]	Permitless Carry	Live Fire	Dangerousness	Suitability	Any Violent Misdemeanor Prohibition for CCW Permit
Tennessee		1996		1989			
Texas		1996		1996			1996
Utah		1995[c]			1995–2010	Pre-1980–2010	1986
Vermont			Pre-1980				
Virginia		1995[c]			1995	Pre-1980–1995	1995
Washington		Pre-1980					
West Virginia		1989–May 24, 2016	2016[e]			Pre-1980–1989	2013–2016
Wisconsin		2011					
Wyoming		1994[c]	2011		1994–2011	1983–1994	

Abbreviation: CCW, concealed-carry weapons.
[a] A dash between years indicates that the law or provision was repealed.
[b] Alaska, Arizona, Arkansas, Illinois, Kansas, Kentucky, Louisiana, Mississippi, Missouri, Nebraska, New Mexico, North Carolina, North Dakota, Ohio, Oklahoma, Tennessee, Texas, Wisconsin, and Wyoming were all considered no-issue CCW states prior to adopting a shall-issue CCW permitting system.
[c] Some limited discretion was afforded to the issuing agency.
[d] Hawaii requires live-fire training for a license to possess a gun, which is a prerequisite for carrying a concealed weapon.
[e] Adoption of the law occurred in the second half of the year during the study period.
[f] Adoption of the law occurred in the first half of the year during the study period.

adoption, 0% of law enforcement agencies reported data in 1988 (the first full year of shall-issue implementation), and 63% of law enforcement agencies reported data in the subsequent 5 years (36). Because the reductions in reporting agency would probably manifest as reductions in reported crime, we elected to remove Florida from our analysis. Kansas passed a stand-your-ground law 1 year prior to adopting a shall-issue CCW law, and Missouri repealed its permit-to-purchase law within 3 years of adopting a shall-issue CCW law, both of which are associated with increases in firearm homicides and thus may present biased estimates of effect related to shall-issue law adoption (37–41). Web Table 1 provides law adoption dates for stand-your-ground and permit-to-purchase laws.

To examine the average effect related to adopting a shall-issue CCW law across states, and to understand whether different shall-issue permit provisions affect the relationship between shall-issue laws and violent crime, we calculated an inverse standard error weighted average using a REMA. We specified a random-effects model to account for difference in implementation and context relevant to gun violence across states and time. Random-effects models assume that there is a distribution of true effect sizes, and the weighted average represents the mean of the population of true effects. For each outcome, we conducted stratified REMA according to the presence of a given CCW permitting provision. We calculated measures of I^2 and T^2 to understand heterogeneity of effects. REMAs have been used previously to understand the average impact of law changes related to firearm violence (6, 42). We used the R package "meta" for this analysis (43).

RESULTS

Figure 1 displays trendlines of fatal and nonfatal outcomes from 1980 to 2019. We provide trend lines for may-issue laws, shall-issue laws, and the national average. For Figure 1, CCW laws are categorized on the basis of whether states are never adopters or adopters. This implies that a late-adopting shall-issue CCW law state is included in the "shall-issue" trendline throughout the duration.

Examination of RMSE performance across model specifications indicated that ASCM+FE provided both the best and most consistent RMSE in the pretreatment period (Figure 2). Figure 2 shows the overall RMSE and the prior-3-years RMSE for all shall-issue–adopting states across all 6 outcomes. Note that, on average, the ASCM+FE models provide the lowest RMSE across model selections and that the gap between the overall RMSE and prior-3-years RMSE is smallest. In addition to RMSE performance, ASCM+FE provided the best covariate balance, on average, comparing treatment and synthetic control models (Web Figures 1–9). Donor pool weights are also provided in Web Figures 10–43. Web Figures 44–77 provide lambda cross-validation for ASCM and ASCM+FE.

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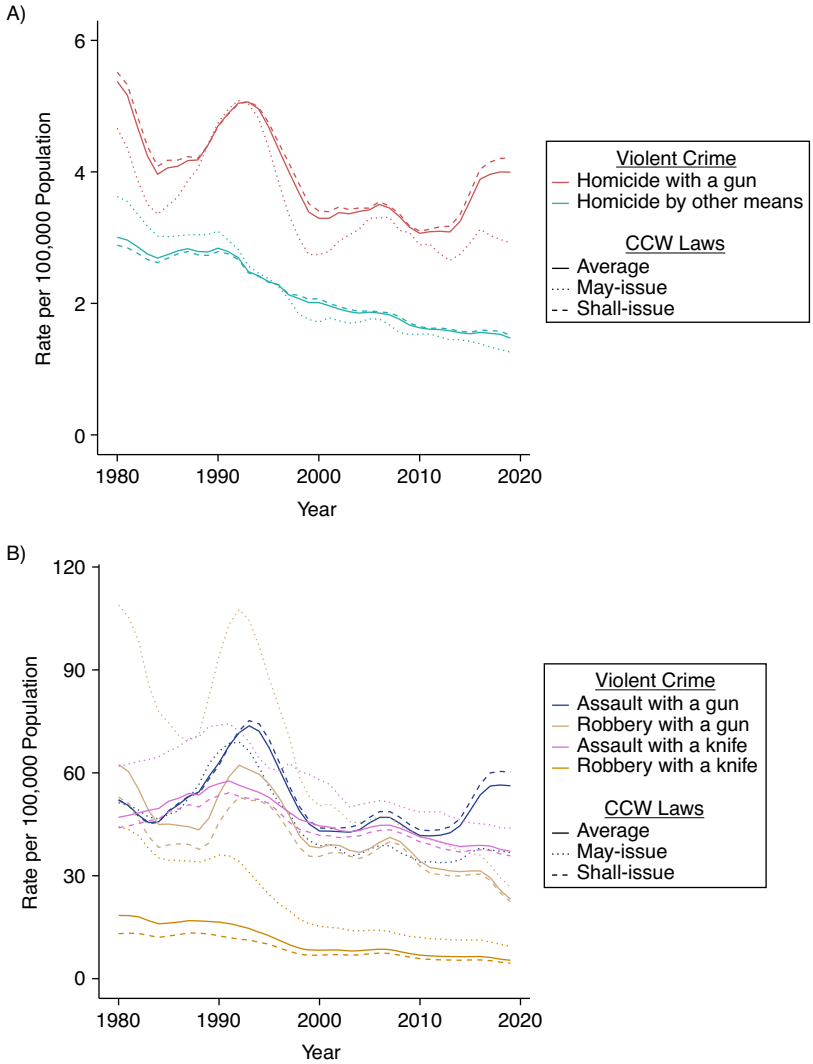

Figure 1. Trends in fatal (A) and nonfatal (B) violent crime rates per 100,000 population according to type of concealed-carry weapons (CCW) law, United States, 1980–2019.

Figure 3 shows a gap plot for the state of Alabama. In the figure, columns represent the 4 model specifications and rows represent the 6 violent crime outcomes. In this image, we see a visual representation of the RMSE for the full prelaw period and the RMSE for the 3 years prior to shall-issue law adoption, as well as the ATT and the standard error. Examining the model specifications for gun homicide rates (row 1), we see that the difference between observed rates and the synthetic control models prior to the shall-issue law was closest to 0 for the 2 models with fixed effects in the pretreatment period. This can also be seen in Figure 2; examining gun homicide in Alabama, we note that the values for ASCM+FE and SCM+FE are clustered together, with the other RMSE values further away from 0. We also see that there is large uncertainty in the non–fixed-effects models (Figure 3). Web Figure 78 shows that the standard errors are much larger for the ASCM and SCM models than for the ASCM+FE and SCM+FE models for gun homicides in Alabama (Web Figures 79–83 show the ATTs and standard errors according to model specifications for the other violent crime outcomes). Gap plots for every treatment state are provided in Web Figures 84–116.

Violent crime, in general, trended upwards during the 10 years after shall-issue law implementation. Table 2 shows the ATTs and standard errors for our violent crime outcomes with an ASCM+FE model specification. Rates of gun and nongun homicide were largely greater in the postimplementation period. Rates of gun assaults were largely increasing, while rates of knife assaults decreased. Rates of gun robbery and knife robbery did not show discernable trends.

Results of the REMA indicated that adopting a shall-issue law was associated with an additional 5.31 gun assaults per 100,000 population (95% confidence interval (CI): 1.19, 9.43) (Figure 4). This translates to a 9.5% increase in rates

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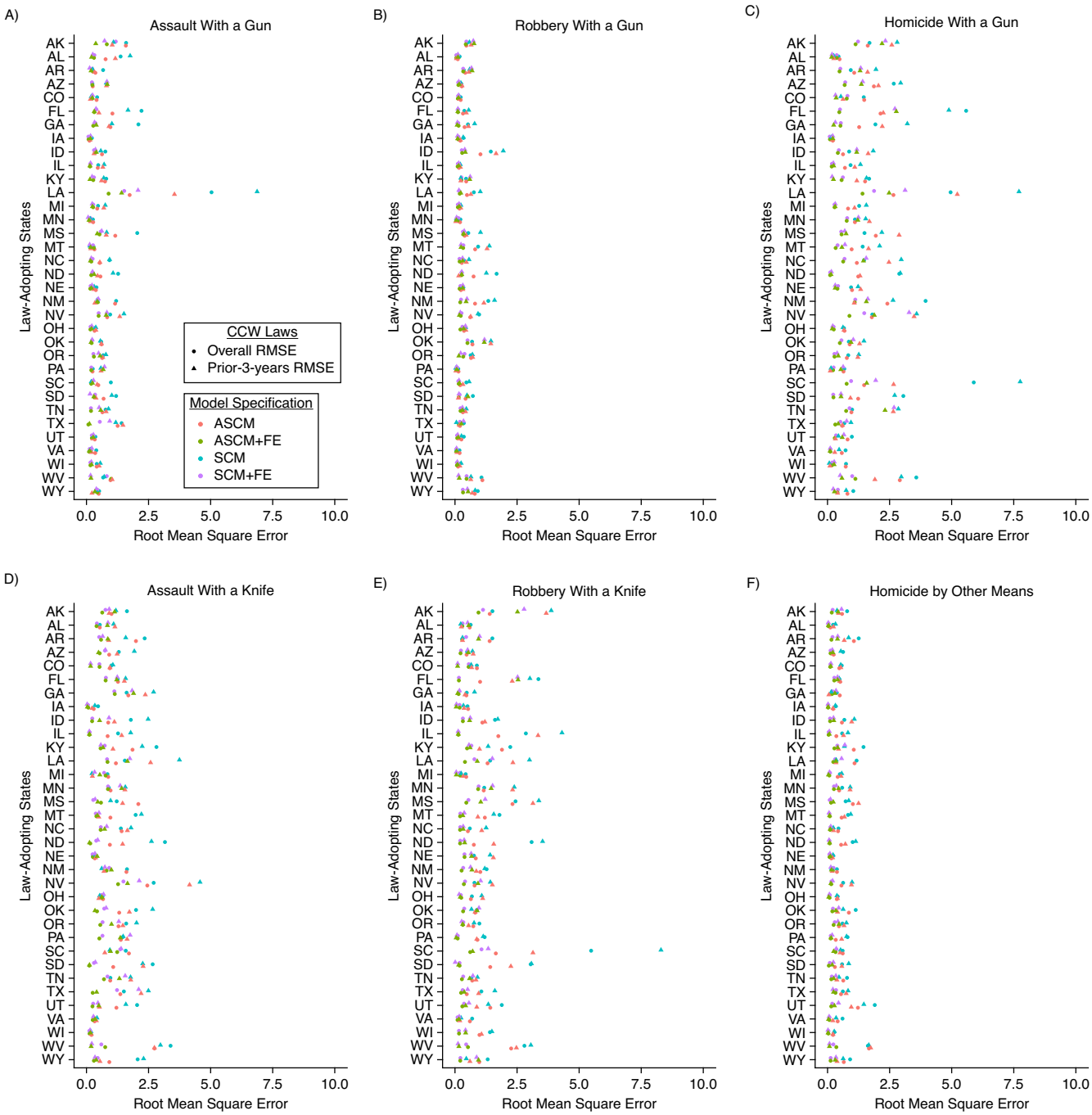

Figure 2. Root mean square errors (RMSEs) from 4 models of the relationship between concealed-carry weapons (CCW) laws and crime rates per 100,000 population for 6 violent crime outcomes, United States, 1980–2019. A) Homicide with a gun; B) homicide by other means; C) assault with a gun; D) robbery with a gun; E) assault with a knife; F) robbery with a knife. Circular points represent the overall RMSE, whereas triangular points represent the RMSE for the 3 years prior to law adoption. Red represents values from augmented synthetic control modeling (ASCM), green represents values from augmented synthetic control modeling with fixed effects (ASCM+FE), teal represents values from synthetic control modeling (SCM), and purple represents values from synthetic control modeling with fixed effects (SCM+FE). AK, Alaska; AL, Alabama; AR, Arkansas; AZ, Arizona; CO, Colorado; FL, Florida; GA, Georgia; IA, Iowa; ID, Idaho; IL, Illinois; KY, Kentucky; LA, Louisiana; MI, Michigan; MN, Minnesota; MS, Mississippi; MT, Montana; NC, North Carolina; ND, North Dakota; NE, Nebraska; NM, New Mexico; NV, Nevada; OH, Ohio; OK, Oklahoma; OR, Oregon; PA, Pennsylvania; SC, South Carolina; SD, South Dakota; TN, Tennessee; TX, Texas; UT, Utah; VA, Virginia; WI, Wisconsin; WV, West Virginia; WY, Wyoming.

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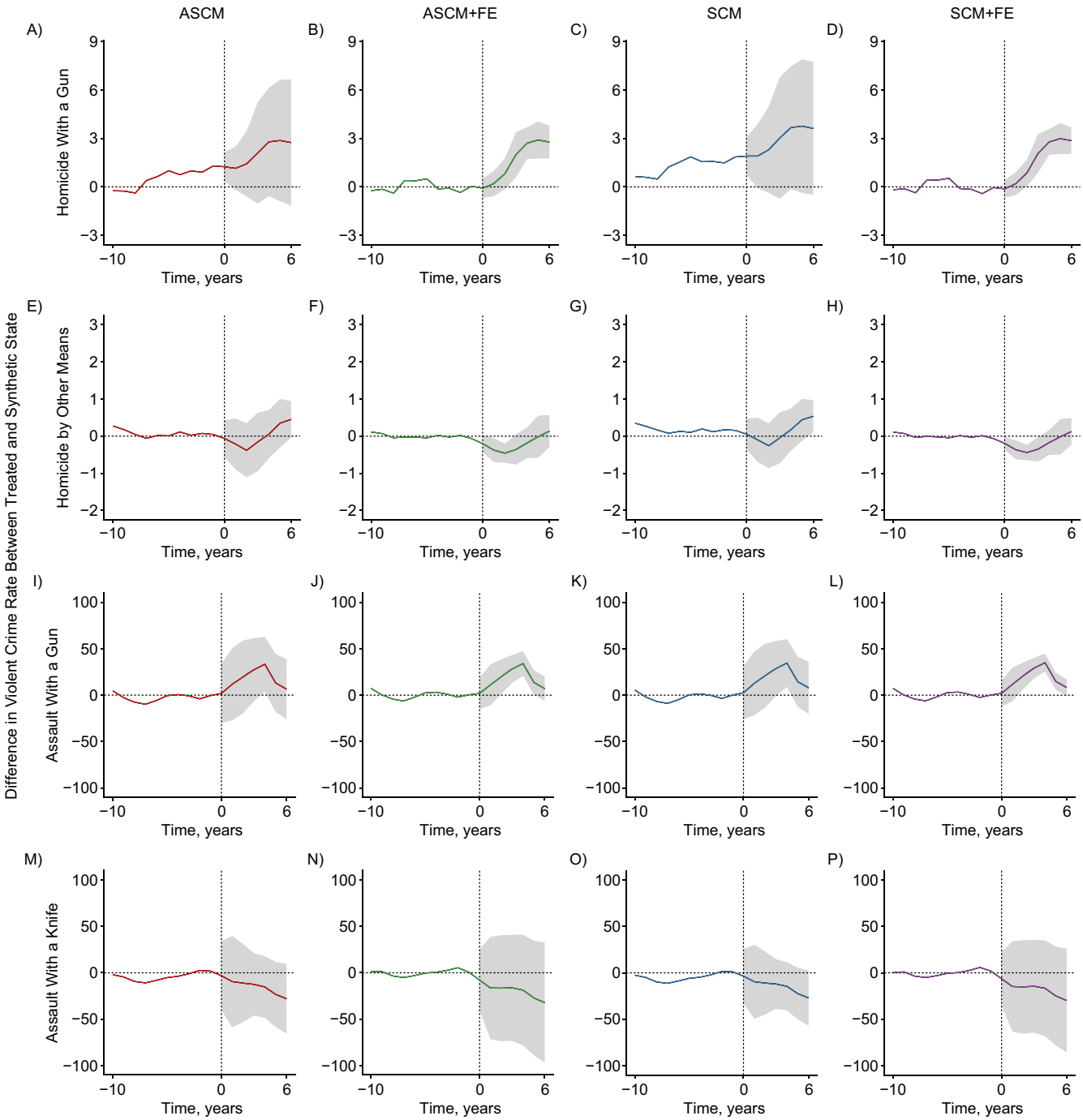

Figure 3 Continues

of gun assaults above forecasted counterfactuals. Shall-issue law adoption was associated with an additional 0.19 nongun homicides per 100,000 population (95% CI: 0.07, 0.31), an 8.8% increase. The adoption of shall-issue laws was associated with non–statistically significant increases in rates of all other violent crimes. Heterogeneity of effect across states was moderate to high for all outcomes except assaults and robberies with a knife (Web Figure 117).

Differential associations related to the presence or absence of shall-issue CCW-law permitting provisions existed by and large for gun assaults and nongun homicides (Web Figures 118–123). For gun assaults, states with violent misdemeanor prohibition provisions saw a nonsignificant increase of 2.81 crimes per 100,000 population (95% CI: −1.51, 7.13). When states allowed violent misdemeanants to obtain CCW permits, shall-issue laws were associated

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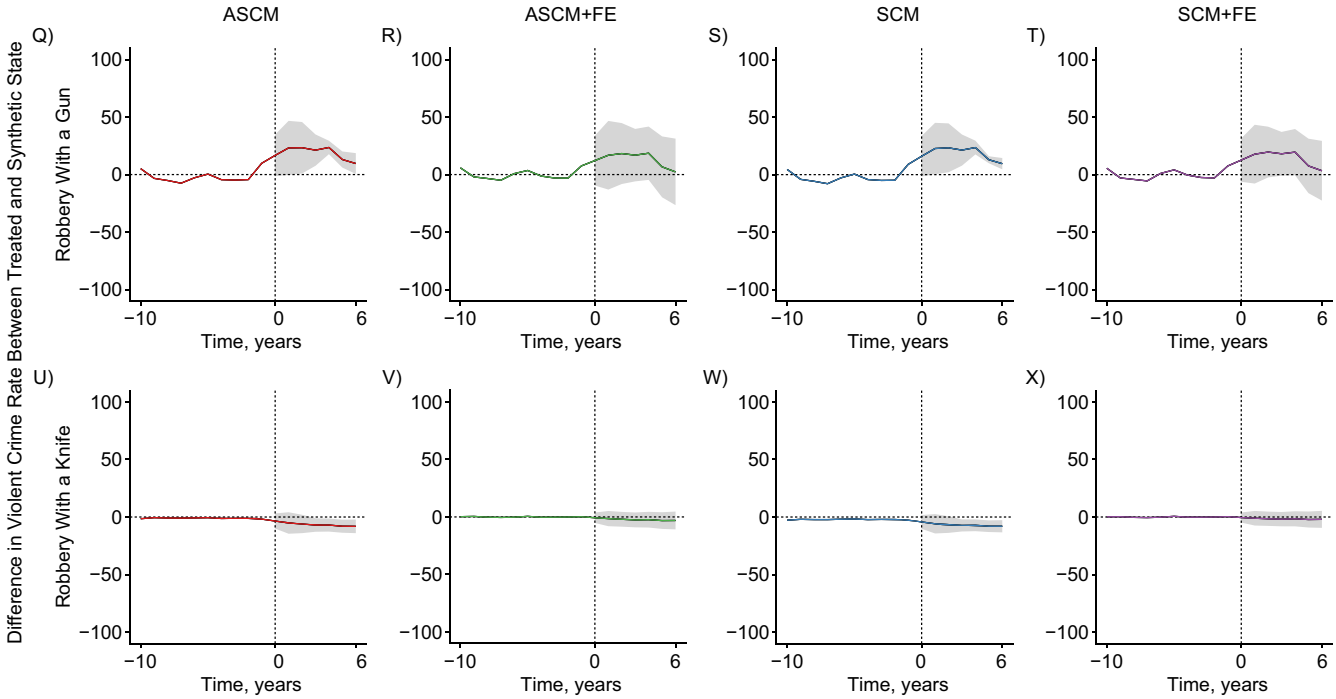

Figure 3. Gap plots comparing treated and synthetic control model outcomes for the relationship between concealed-carry weapons laws and rates of 6 violent crimes, Alabama, 1980–2019. The 4 columns of data represent crime rates per 100,000 population for augmented synthetic control modeling (ASCM) (red), augmented synthetic control modeling with fixed effects (ASCM+FE) (green), synthetic control modeling (SCM) (teal), and synthetic control modeling with fixed effects (SCM+FE) (purple), respectively. Plots A–D, homicide with a gun; plots E–H, homicide by other means; plots I–L, assault with a gun; plots M–P, assault with a knife; plots Q–T, robbery with a gun; plots U–X, robbery with a knife.

with a significant increase of 12.7 gun assaults per 100,000 population (95% CI: 0.97, 24.42). Shall-issue states with live-fire training requirements and suitability provisions displayed nonsignificant relationships with gun assaults, while states without such provisions displayed significantly greater rates. The same differential relationships existed for nongun homicides.

Web Figures 118–123 also present differences in estimates of shall-issue law effects based on the number of provisions intended to reduce risks. States that adopted shall-issue CCW laws without any provisions saw 10.26 (95% CI: 2.87, 17.65) additional gun assaults and 1.44 (95% CI: 0.44, 2.45) additional gun homicides per 100,000 population—21.6% and 29.8% increases in rates above those expected, respectively. Differential associations also existed for shall-issue CCW law states with at least 2 and at least 3 permitting provisions for rates of gun assault and nongun homicide. For both outcomes, shall-issue states with at least 2 or at least 3 provisions displayed nonsignificant relationships, whereas states without at least 2 or without at least 3 provisions displayed significant increases in association.

DISCUSSION

There have been dramatic changes in state laws governing civilian carrying of concealed firearms from the 1980s to 2021, resulting in large-scale deregulation. Consistent with many recent studies (5–8, 14), we found that states that moved from a may-issue CCW law to a less restrictive shall-issue CCW law experienced increased rates of violent crime. Specifically, shall-issue law adoption was associated with a 9.5% increase in rates of aggravated gun assault. Shall-issue laws were also associated with an 8.8% increase in nongun homicides.

Evidence of violence-generating effects of moving from may-issue laws to shall-issue laws was most evident when states allowed individuals with prior convictions for a violent misdemeanor crime to obtain CCW permits. Under these conditions, shall-issue laws were associated with significant increases in gun assault rates. Shall-issue laws that prohibited violent misdemeanants from obtaining CCW permits were not associated with changes in gun-related violent crime. Violent misdemeanants who are allowed to purchase handguns legally subsequently commit violent and firearm-related crimes at 7 times the rate at which handgun purchasers without such convictions commit these crimes after purchasing a handgun (43). Laws that prohibit violent misdemeanants from acquiring firearms are associated with lower rates of violent and firearm-related crime (44). State-level studies have also found firearm prohibitions for violent misdemeanants to be associated with reduced rates of intimate partner homicides (45), total homicides (12), and firearm injuries treated in hospitals (46).

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Table 2. Average Treatment Effect in the Treated[a] (and Standard Error[b]) for the Association Between Adoption of a Shall-Issue Concealed-Carry Weapons Law and Violent Crime Outcomes, United States, 1980–2019

State	Homicide Committed With a Gun	Homicide Committed by Other Means	Assault Committed With a Gun	Assault Committed With a Knife	Robbery Committed With a Gun	Robbery Committed With a Knife
Alabama	1.62 (0.43)[c]	−0.21 (0.18)	16.56 (7.87)[c]	−19.26 (27.70)	13.10 (12.49)	−2.19 (3.20)
Alaska	−0.20 (0.50)	1.22 (0.64)	8.12 (21.48)	0.96 (17.30)	−8.95 (25.38)	−0.06 (6.09)
Arkansas	0.34 (1.11)	0.82 (0.74)	23.90 (23.17)	−3.00 (19.03)	17.06 (38.14)	8.73 (12.99)
Arizona	2.44 (0.60)[c]	0.83 (0.22)[c]	−1.60 (17.75)	−8.91 (7.06)	1.69 (22.59)	6.73 (6.46)
Colorado	−0.98 (0.61)	0.04 (0.14)	15.32 (9.30)	−0.10 (6.03)	−2.64 (9.04)	−2.78 (3.37)
Florida	−3.10 (0.39)[c]	−0.41 (0.27)				
Georgia	−0.96 (0.82)	−0.43 (0.19)[c]	4.52 (8.42)	−6.32 (12.12)	7.54 (13.17)	−2.25 (1.97)
Iowa	0.42 (0.19)[c]	0.19 (0.11)	8.34 (0.49)[c]	0.57 (14.68)	7.21 (9.14)	1.19 (1.39)
Idaho	0.33 (0.41)	0.50 (0.26)	−1.78 (17.02)	−0.61 (23.38)	−19.36 (25.68)	3.06 (3.79)
Illinois	1.50 (0.20)[c]	−0.06 (0.06)	−3.28 (5.68)	0.87 (8.89)	5.59 (8.66)	−0.76 (2.37)
Kentucky	−0.06 (0.89)	1.05 (0.39)[c]	4.25 (16.34)	−16.15 (24.01)	7.96 (23.12)	2.72 (6.26)
Louisiana	0.61 (2.43)	0.52 (0.32)	6.30 (15.04)	25.17 (35.77)	12.40 (31.22)	4.26 (13.92)
Michigan	−0.70 (0.86)	−0.31 (0.12)[c]	2.80 (9.70)	−2.53 (8.63)	1.76 (9.19)	−2.02 (3.68)
Minnesota	−0.42 (0.74)	−0.11 (0.21)	−0.38 (11.05)	−11.56 (14.29)	−0.13 (10.55)	−2.79 (3.48)
Mississippi	1.39 (0.67)[c]	1.22 (0.38)[c]	4.16 (18.99)	−13.64 (20.14)	13.02 (25.71)	13.02 (7.09)
Montana	−0.10 (0.30)	0.85 (0.46)	−17.70 (14.99)	−15.49 (18.74)	−18.32 (21.53)	2.33 (4.03)
North Carolina	−0.15 (1.06)	0.06 (0.37)	22.15 (23.64)	−17.61 (18.80)	23.88 (28.56)	0.43 (10.52)
North Dakota	−0.01 (0.40)	0.03 (0.14)	−2.33 (7.20)	−8.96 (12.43)	9.48 (13.27)	0.62 (7.14)
Nebraska	0.12 (0.43)	0.11 (0.08)	4.28 (9.34)	−1.33 (9.73)	−2.42 (1.59)	−0.75 (0.35)[c]
New Mexico	−0.42 (1.21)	−0.22 (0.18)	−10.38 (4.16)[c]	−9.08 (11.47)	−10.29 (3.68)	−1.53 (1.64)
Nevada	1.56 (1.07)	0.18 (0.24)	8.90 (36.74)[c]	16.29 (9.21)	−3.43 (29.01)	−6.71 (8.84)
Ohio	0.41 (0.76)	0.41 (0.21)	8.66 (2.72)[c]	0.31 (5.56)	11.60 (4.02)[c]	0.38 (1.64)
Oklahoma	−0.33 (0.48)	0.40 (0.29)	11.71 (19.50)	7.44 (10.96)	−14.47 (31.71)	4.43 (7.28)
Oregon	−0.01 (0.35)	0.19 (0.29)	−6.34 (6.24)	−5.84 (13.68)	−31.58 (21.38)	−5.35 (5.67)
Pennsylvania	1.65 (0.96)	0.33 (0.13)[c]	20.68 (17.02)	8.89 (6.27)	36.67 (10.82)	5.76 (2.73)
South Carolina	0.78 (0.70)	0.30 (0.27)	8.51 (24.75)	−27.58 (38.45)	9.61 (44.87)	−3.14 (11.67)
South Dakota	−0.46 (0.50)	−0.63 (1.02)	9.31 (9.85)	1.89 (8.92)	7.68 (15.75)	−1.38 (8.55)
Tennessee	0.58 (0.70)	0.33 (0.37)	73.39 (14.34)	43.02 (11.91)	13.39 (20.98)	5.47 (4.11)
Texas	−0.40 (2.98)	0.19 (0.10)	10.17 (16.21)	−0.99 (12.94)	−3.00 (48.64)	11.46 (8.45)
Utah	0.24 (0.60)	0.54 (0.26)	7.97 (16.37)	10.21 (11.87)	6.83 (14.21)	6.83 (4.33)
Virginia	−0.55 (1.23)	0.46 (0.17)[c]	−2.32 (27.95)	−11.18 (14.35)	−5.66 (39.48)	−0.70 (7.23)
Wisconsin	0.83 (0.23)[c]	−0.01 (0.08)	13.63 (4.31)[c]	17.40 (13.58)	7.97 (9.85)	1.65 (1.19)
West Virginia	−0.48 (1.21)	0.96 (0.40)[c]	−1.88 (9.76)	2.12 (14.57)	−4.78 (24.88)	8.12 (3.73)
Wyoming	0.02 (0.47)	1.22 (0.29)[c]	−22.50 (21.93)	−16.65 (17.07)	−28.23 (36.88)	−1.40 (12.16)

[a] The augmented synthetic control method with fixed effects was used for model specifications. Each model also included jackknife standard errors for inference.

[b] All standard errors were calculated using the jackknife method. Web material shows the pretreatment covariate balance (Web Figures 1–9), donor pool contribution (Web Figures 10–43), and gap plots (Web Figures 84–116) for each of these synthetic control models.

[c] The 95% confidence interval did not include 0.

Assaults are the most common type of violent crime. Increasing gun-carrying could increase the frequency with which individuals use firearms inappropriately in response to interpersonal conflicts, provocations, or situations that they want to control by threat of deadly force. Berkowitz and Lepage (47) found evidence of a "weapon effect" (48) linking the presence of a firearm with increased aggressive thoughts, perceptions of situations as hostile or dangerous, and aggressive behavior. Most legal gun owners may be law-abiding and not prone to violence; however, a small percentage may be at heightened risk of committing violent acts. For example, between 2001 and 2003, the National Comorbidity

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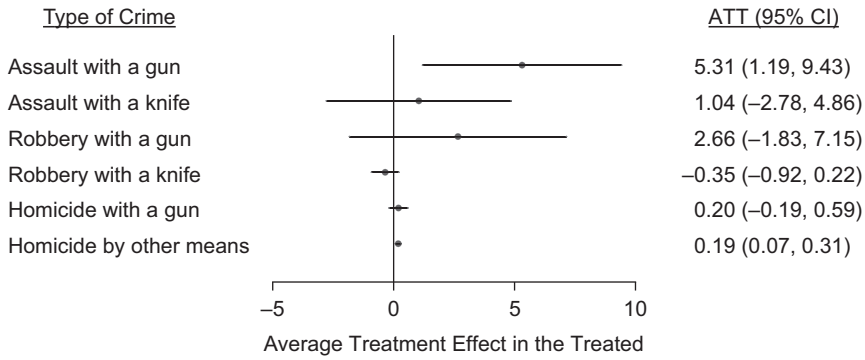

Figure 4. Impact of adoption of shall-issue concealed-carry weapons laws on violent crime outcomes in a random-effects meta-analysis, United States, 1980–2019. Bars show 95% confidence intervals (CIs). The average treatment effect in the treated (ATT) represents the change in the number of crimes per 100,000 population.

Survey Replication collected extensive data on a nationally representative sample of adults. Swanson et al. (49) found that nearly 1 in 20 men surveyed reported carrying a gun outside the home and impulsive, angry behavior patterns.

As cultural norms regarding civilian gun-carrying shift after shall-issue law adoption toward more individuals' being armed with a concealed firearm, more hostile altercations are likely to involve firearms, such as an argument over a parking space (50, 51). There is also evidence that drivers drive more aggressively when there is a gun in the vehicle (52, 53), and one news report noted an uptick in road rage incidents involving fatal shootings in recent years (54). In a systematic review, Braga et al. (55) synthesized existing literature on the instrumentality of firearms. They found that the presence of firearms likely made otherwise nonfatal altercations result in a fatality (55). Further, recent evidence from Donohue et al. (56) found that adopting a right-to-carry law was associated with a 35% increase in firearm theft—meaning that adoption of these laws may introduce thousands of guns into illegal gun markets and/or the hands of criminals. Given this lethality, increased exposure to firearms outside the home in response to legal and cultural changes supporting more gun-carrying appears to increase the likelihood of violence involving firearms.

Opponents of firearm regulations often argue that legal gun owners prevent violent crime, especially if they are allowed to carry concealed guns in public. However, many states have relatively low standards for legal gun possession. In those states, the greatest numbers of persons incarcerated for firearm-related violent crimes were persons who legally possessed a gun before using that gun to commit the crime leading to their incarceration (57).

Shall-issue laws were associated with increases in nongun homicides, but we did not find a significant link between these laws and rates of knife assaults and knife robberies. Many nongun homicides are due to domestic violence involving family members or intimate partners and usually occur in the home. The association between shall-issue laws and these crimes is most likely due to confounding with conditions more specific to domestic homicide than to factors influencing violence outside the home, given the null findings and point estimates near 0 in the aggregate models for assaults and robberies committed with knives. The strongest and most consistent association for shall-issue laws was an increase in aggravated gun assaults.

Stratifying our REMA analysis by the absence or presence of shall-issue permitting provisions produced differential results. For aggravated gun assaults and nongun homicides, states with shall-issue laws that required the permittee to undergo live-fire training, allowed law enforcement officers limited discretion regarding a permittee's suitability to carry a concealed weapon, and/or prohibited violent misdemeanants from obtaining a permit did not experience increased rates of violent crime in comparison with states without such provisions. The cumulative impact of these provisions was notable; states with shall-issue laws that had at least 1, 2, or 3 of the provisions we examined had no statistically significant relationship with rates of gun assault or nongun homicide.

This study was not without limitations. Like other methods for estimating the impact of public policies, the ASCM and SCM methods rest on several assumptions. The methods' ability to provide an accurate counterfactual on which to make causal inference is based on the pretreatment fit between treated states and their ASCMs. A strong pretreatment fit is necessary to fulfill the parallel trend assumption needed for causal inference. We assessed and reported not only overall model fit but model fit for the 3-year period leading up to shall-issue-law adoption to identify potential biases. The literature is currently uncertain as to what indicates "good" pretreatment fit. We defined strongly performing models as having a pretreatment RMSE within 2 standard deviations of the mean RMSE and calculated our inverse-standard-error–weighted averages excluding states with poor treatment fit.

ASCM assumes no anticipation in effect and no spillover from the donor pool. To examine no anticipation in effect, we calculated the prior-3-years RMSE and compared it with the overall RMSE as part of our model selection. Because of our very small donor pool ($n = 8$), we had limited ability to test the assumption of no spillover. It is likely that a citizen in a shall-issue state with a CCW license could have carried

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their weapon unlawfully into a may-issue state—although, to affect our model estimates, they would have needed to commit a violent firearm-related crime. The likelihood that this scenario happened frequently enough to impact state-level rates of violent crime per 100,000 population is small.

The synthetic control method is also limited by its inability to control for other state laws. As such, we excluded Missouri and Kansas as treatment states a priori. We also could not assess implementation and enforcement of law provisions. Training practices or limited law enforcement discretion, for example, may not be implemented in a way that reflects statutory text. Future work should examine specific permitting standards using quantitative and qualitative methods. However, our use of the REMA method did allow us to understand whether the presence of certain CCW-law permitting provisions confounded the relationship between law adoption and violent crime rates.

Use of the Federal Bureau of Investigation's "Offenses Known and Clearances by Arrest" data had limitations. The primary limitation is that this data set only includes crimes reported to police, and many crimes go unreported. A 2008 National Crime Victimization Survey found that the percentages of personal crimes reported to police were highest for aggravated assault with an injury (74.1%) and robbery with an injury (72.7%) (58). Thus, the UCR Program is likely to account for the most serious forms of violent crime. Reporting data to the UCR Program is done on a voluntary basis, and reporting practices may vary by state and over time (59). The National Archive of Criminal Justice Data, where these data are archived, imputes county-level data when there is missingness and the imputed values are aggregated at the state level (60). Imputation for problems with state-level UCR data that are flagged on the UCR "Methods" page reveal no connection between such problems and the states and years of shall-issue adoption (61). The limitations surrounding underreporting and missingness in UCR data probably contribute to measurement error, and the direction of bias due to nondifferential misclassification is hard to predict.

In conclusion, during the 10 years of postadoption time, adoption of shall-issue CCW laws was associated with increased rates of aggravated gun assault and nongun homicide. It appears that restricting shall-issue CCW permits from petitioners with violent misdemeanor convictions mitigates some of the harmful impact of adopting a shall-issue law, as does requiring live firearm training. Shall-issue states should consider adopting a violent misdemeanor prohibition or a live firearm training provision as part of their CCW system. States that become shall-issue states in the future should work to ensure that these provisions are included in their CCW systems to lessen the negative health impacts of the adoption of such laws.

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